SOLUTIONS

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SOLUTE:

A substance that dissolves in a solvent is called a solute.

SOLVENT

A substance that has the ability to disperse a solute is called a solvent.

SOLUTION

A homogenous mixture or single phase mixture of the molecules, atoms or ions of two or more components is called a **Solution**.

TYPES OF SOLUTION

TRUE SOLUTION

- A solution in which particle size of the solute is less than 1 µm. For example, salt solution.

COLLOIDAL SOLUTION

A solution in which particle size is more than 1 µm (1-200 µm) and the particles are dispersed in the solution. For example, protein and polysaccharide solutions, albumin.

CRYSTALLOID SOLUTION

- A solution of low molecular weight ions which can pass through a semi-permeable membrane is called a crystalloid e.g. solutionRinger's lactate

EMULSION

- A mixture of two immiscible liquids is called an emulsion. For example, oil in water.

PHARMACEUTICAL IMPORTANCE:

- Taste and smell of the oil is masked.
- Oil becomes easily digestible.
- There is uniform distribution of the drug.

SUSPENSION

- A mixture in which the particle size of the solute is greater than 200 mμ which are suspended in the solution is called a suspension. Particles are visible with the naked eye. Example is mucaine.

PHARMACEUTICAL IMPORTANCE:

- Suspension is ideal dosage form for patients who have difficulty in swallowing tablets or capsules, as in case of children and elderly patients.
- Drugs are absorbed more rapidly from a suspension than from tablets or capsules.
- Disagreeable taste of the drug can be masked in a suspension form. E.g. chloramphenicol palmitate.
- Suspension is more stable than solutions.

PREPARATION OF SOLUTIONS

W/W % SOLUTION:

- A weight per unit weight solution means number of grams of solute per 100gm solution. 5% w/w solution means 5gm solute and 95 gms solvent.

W/V % SOLUTION

- A weight per unit volume solution means number of grams of solute per 100 ml of solution. 5% w/v solution means 5 gm solute in 100 ml of solution.

V/V % SOLUTION

- A volume per unit volume solution means number of ml of solute per 100 ml of solution. 5% v/v solution means 5 ml of solute and 95 ml of solvent, i.e.100 ml solution.

ATOMIC WEIGHT

- Sum of the number of neutrons and protons in the nucleus of an atom is called its atomic weight

MOLECULAR WEIGHT

- Sum of atomic weights of atoms present in a molecule

EQUIVALENT WEIGHT

- Gram molecular weight divided by the valency

$$\text{Equivalent Weight} = \frac{\text{molecular weight (or molar mass)}}{\text{Number of equivalents}}$$

MOLE

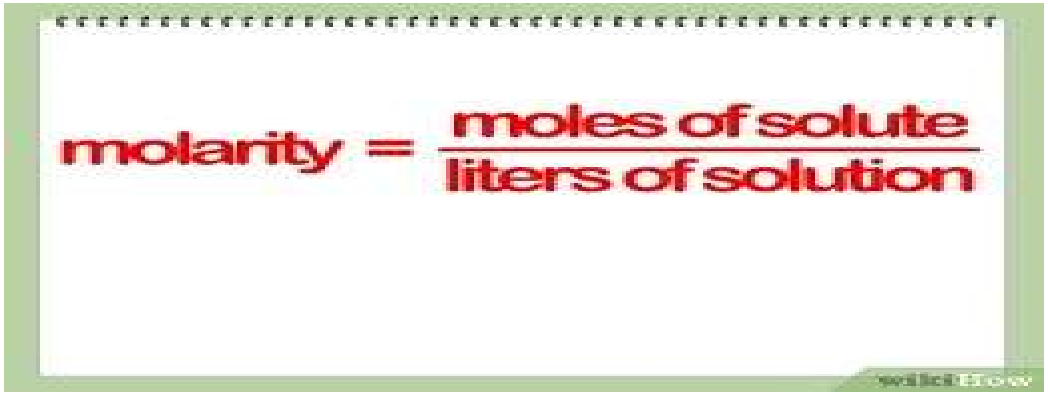
- Gram molecular weight .

OSMOLE

- Any substance/molecule/ion that can exert osmotic pressure .

MOLARITY

- The number of moles of solute in one litre or 1000 ml of solution.



MOLALITY

- The number of moles of solute per kg of solvent.

$$\textbf{Molality} = m = \frac{\textbf{moles of solute}}{\textbf{kg of solvent}}$$

$$m = \frac{mol}{kg}$$

NORMALITY

- The gram equivalent weight of solute in one litre or 1000 ml of solution.

$$Normality(N) = \frac{gram\ equivalent\ weight}{liter\ of\ solution}$$

OSMOLALITY

- The number of particles of solute per unit weight of water irrespective of the size and nature of particles. Units are mmol/kg of water.

OSMOLARITY

- The number of particles of solute per litre of solution. Units are mmol/L.

Osmolarity Formula

$$\text{Osmolarity} = \frac{\text{moles ions}}{\text{L solution}}$$